Carisurg

Assignment 4

Report

Focus Topic: SpO2 —Oxygen Saturation.

Due Date: May 25th, 2025.

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Oxygen Saturation (SpO_2)

Having worked with FiO_2 — the fraction of inspired oxygen — in our dataset, it became apparent that its most clinically meaningful partner was absent. FiO_2 tells you what oxygen concentration a patient is receiving, but without SpO_2, you have no way of knowing how effectively their body is actually using it. It felt only natural to write about SpO_2 as the missing half of that picture.

Oxygen Saturation (SpO_2) is a measure of the percentage of haemoglobin in the blood that is carrying oxygen, typically measured at triage using a pulse oximeter placed on the fingertip. It represents how effectively the lungs are transferring oxygen into the bloodstream and is expressed as a percentage.

The normal clinical range for SpO_2 in a healthy adult is **95–100%**.

A reading **below 95%** demands clinical attention, and a reading **below 90%** is considered **hypoxaemia** — a state of dangerously low blood oxygen that requires immediate intervention.

In patients with chronic respiratory conditions such as COPD—Chronic Obstructive Pulmonary Disease, a baseline SpO_2 of **88–92%** may be their normal, which is an important nuance for triage nurses to account for when interpreting readings.

In an emergency department triage setting, SpO_2 is one of the fastest and least invasive vital signs to obtain. A drop in SpO_2 can precede visible signs of respiratory distress, making it an early warning indicator that a patient is deteriorating before they appear overtly unwell.

When read alongside FiO2, the picture becomes even more powerful — a patient on high-flow oxygen (high FiO2) with a low SpO2 is in far more danger than their oxygen delivery alone would suggest, as it indicates their lungs are failing to utilise the oxygen being provided.

At Mercer General, capturing SpO2 routinely alongside FiO2 would significantly strengthen the dataset's ability to identify patients at respiratory risk early in the triage pathway.